

F = ma Exam FAQ

Everything about *taking* the exam — the logistics, not the physics

For official information, use the AAPT (American Association of Physics Teachers) website. Trust that website above anything in this handout.

www.aapt.org/physicsteam

The two pages worth bookmarking are [Registration](#) and [Rules](#).

1 The Short Version

The **F = ma exam is 25 multiple-choice questions in 75 minutes, on mechanics only. No calculus is required.** You take it **in person, on a computer, with a live proctor watching**, on a single day in February. **Your school has to register you** — you cannot sign yourself up — and registration closes about three weeks before the exam. Score high enough (recently 14–18 out of 25) and you are invited to the USAPhO exam in April.

2 Timeline: What To Do, and When

When	What you should be doing
September–October	Ask your physics teacher whether the school will run the F = ma exam. If they have never heard of it, send them the AAPT link. Don't leave this until January. If your school says no, you can ask teachers at other schools — you do not have to test at your own.
December–early January	Make sure the proctor has actually registered. Do not assume. Ask them to confirm they have submitted it.
Mid-January	Registration deadline. In 2026 this was January 20 (extended to January 21). Once it passes, there is no way in for that year.
Mid-February	Exam day. In 2026 it was February 12, in a 1:00–4:00 pm EST window — your proctor picks the start time inside that window. In 2024 and 2023 it was February 8 and 9.
Early March	Solutions posted (March 5 in 2026), then results (March 9 in 2026).
April	USAPhO , if you qualified. April 10 in 2026, a 1:00–5:00 pm EDT window.
June	Top scorers on the USAPhO are invited to the U.S. Physics Team training camp (May 30–June 9, 2026, at the University of Maryland).

3 The Exam Itself

What is on it?

Mechanics only — kinematics, Newton's laws, energy, momentum, rotation, oscillations, gravitation, fluids. No electricity and magnetism, no thermodynamics, no optics, no modern physics.

How long is it and how many questions?

25 multiple-choice questions in 75 minutes. That is three minutes per question on average.

Do I need calculus?

No. Every question can be done without it. Calculus occasionally makes a problem faster, but nothing on the exam requires it.

How is it scored? Is there a guessing penalty?

One point for each correct answer, and **zero points for a wrong answer or a blank**. AAPT's own instructions say "There is no additional penalty for incorrect answers."

So never leave a question blank. A blank and a wrong answer score identically, and a guess among five choices is worth 0.2 points on average. If you have eliminated even one option, guess. With two minutes left, put an answer on everything you have not reached.

Are the questions in order of difficulty?

Roughly, but not reliably. AAPT states that "All questions are equally weighted, but are not necessarily equally difficult." Since every question is worth the same one point, there is no reason to grind on question 8 while questions 20 and 21 — which you might find easier — go unread. Take a pass through the whole exam.

What value of g should I use?

The exam tells you: use $g = 10 \text{ N/kg}$ unless a question says otherwise. This is deliberate — it keeps the arithmetic clean.

4 How You Actually Take It

Is it on paper or on a computer?

On a computer. As of the 2026 cycle, AAPT states that the F = ma and USAPhO exams are "conducted online via Educational Vistas only," and that "There will be no printable versions of the exams available." This changed recently — older exams were paper with a bubble sheet, and in 2021 the platform was Art of Problem Solving — so ignore what you read in older guides.

Can I take it at home?

No. It is proctored and taken live, in person, at your school or another approved site, with everyone at that site testing at the same time. It is not a take-home or an open-window online test.

Do I get scratch paper?

Yes, and only from the proctor. The rule is that “Only scratch paper provided by the proctors may be used,” and “The proctors must collect it immediately after the exam.” So you cannot bring your own paper, and you do not get your work back.

What are the calculator rules?

Bring your own scientific calculator. The proctor won’t provide one. You cannot use a graphing calculator like the TI-84.

What else is banned?

Cell phones may not be used during the exam or while the exam materials are present, and you may not use any external references — no books, no formula sheets, no notes. There is no formula sheet provided either.

Can I talk about the questions afterward?

Not right away. Each exam carries an embargo: the 2025 exam told students not to communicate anything about the questions or their solutions until after February 13, 2025 — the day after the exam. Solutions are posted publicly a few weeks later.

5 Getting a Seat: Proctors, Registration, and Fees

Who signs me up?

Your school registers you. AAPT says “It is preferred that one person register all students from a school,” and that person is normally the teacher who will proctor. Your job is to find that person and make sure they do it before the deadline.

Who is allowed to proctor?

- A proctor “should have at least a 2-year degree, though this degree does not need to be related to physics.” Your English teacher can proctor.
- **Not** a parent, relative, or close friend or acquaintance of a student taking the exam.
- **Not** a for-profit test prep center or educational service — AAPT specifically disallows this now.
- They should register with an official school email address where possible.

What does it cost?

In 2026: **\$75 per school** (\$37.50 if the registering teacher is an AAPT member) **plus \$15.50 per student.**

Do I have to pay it?

Often not. **Many schools cover the fee** out of a department or activities budget, especially if a few students sign up together. Ask — do not assume you will be billed.

Recruit a friend or two. It makes the per-student cost lower, it makes the school far more likely to say yes, and it gives the teacher a reason to run it. “Can you proctor an exam for five of us?” is a much easier ask than “Can you proctor an exam for me?”

My school says no, or has no physics teacher. Now what?

You are not stuck. You need a proctor and a site, and it does not have to be your own school:

- **Another school.** A nearby high school already running the exam can usually add one more student. This is common and entirely allowed.
- **A community college or university.** Many have a testing center.
- **A public library.** Some offer proctoring services.

AAPT’s advice to homeschooled students is exactly this — contact community colleges, universities, or libraries. The same route works for anyone whose school will not host it.

Will AAPT find me a proctor or a test center?

No. AAPT is explicit that it “does not facilitate or arrange test centers, teachers, or proctors,” and does not keep a list you can consult. Finding a site is on you. Start early and ask several places at once.

Am I eligible?

You must be a U.S. citizen, a U.S. permanent resident (green card holder), or currently attending a U.S. school — **and you must be physically located in the U.S. when you take it.** There is no grade-level or age restriction on the $F = ma$ exam itself. Age limits apply only later, to the training camp and the international team.

Can I take it more than once in a year?

Not currently. There is one exam per year. In 2022 there were two (an A and a B version) and students could take both, but every year since has had a single exam. If you miss it, you wait a year.

6 What Score Do I Need?

What is the cutoff?

There is no fixed cutoff. AAPT sets it *after* the exam, at whatever score invites roughly 400–500 students to the USAPhO. So it moves every year with the difficulty of the paper. Here is the actual history:

Year	Exam date	Took it	Median	Mean	Cutoff	Qualified
2021	Feb 18	3524	8	8.89	15	—
2022 (A)	Feb 9	2038	8	8.24	15	—
2022 (B)	Feb 15	2518	8	8.51	15	—
2023	Feb 9	5173	10	10.29	18	417 (8.1%)
2024	Feb 8	6061	7	7.93	14	500 (8.2%)
2025	Feb 12	5848	7	8.21	15	518 (8.9%)
2026	Feb 12	6682	8	8.90	16	440 (6.6%)

Source: AAPT’s official “*F = ma Contest Results*” documents for each year.

So what should I actually aim for?

Aim for 18 or higher on recent exams. That clears every cutoff of the last several years with room for a bad day.

One caveat when you practice with old exams: **the older exams are somewhat easier.** On an exam more than about ten years old you should expect to score higher than 18 to demonstrate the same level of performance that an 18 represents on a current exam. Don’t read a strong score on a 2011 exam as evidence you are already there.

It is also worth reading the medians in the table: 7 to 10 out of 25. **Half of everyone who takes this exam scores single digits**, and those are self-selected students who chose to take a physics olympiad qualifier. Qualifying means finishing in roughly the top 7–9% of that group. A score you might think of as mediocre — 12, 13 — is well above the national median. Do not measure yourself against 25.

Does my score get published?

No. Individual scores are not disclosed publicly. AAPT publishes the aggregate statistics and the cutoff, and notifies qualifiers.

7 After the Exam

What happens if I qualify?

You are invited to the **USAPhO** — a much longer free-response exam, also proctored, held in April (April 10 in 2026, scheduled in a 1:00–5:00 pm EDT window). It is a completely different animal: a handful of long multi-part problems where you show all your work, not multiple choice. Check AAPT for the exact length and structure in the year you take it.

And after that?

Top scorers on the USAPhO are invited to the training camp at the University of Maryland in late May, and five students are eventually selected for the U.S. team at the International Physics Olympiad. Some years AAPT also runs an additional exam, the USAPhO Plus, to help make that selection.

What if I do not qualify?

It’s normal to take the exam more than once before moving on to the USAPhO. The exam is offered every year, the scores in the table above tell you how selective it is, and the ordinary path

is to take it, see where you stand, and come back stronger. That is what the practice tests and the handouts are for.

8 Your Checklist

- ☐ Asked a teacher about proctoring (**do this in the fall**)
- ☐ Confirmed who is registering and whether the school pays
- ☐ Recruited a friend or two to bring the cost down
- ☐ Confirmed the proctor actually submitted the registration
- ☐ Checked the exam date and start time with the proctor
- ☐ Bought a non-graphing scientific calculator and cleared its memory
- ☐ Know that blanks score zero, so make your best educated guess on any problems you haven't solved

Questions this handout does not answer? Ask me — and check aapt.org/physicsteam, which is always the final authority.